

# The impact of social networking sites on the students around Kolkata

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## OBJECTIVE OF THE PROJECT

### A. Description of the problem:

The prevalence of social networking sites (SNS) has significantly increased over the past decade, reshaping how students interact and communicate. The rise of platforms like Facebook, Twitter, and Instagram has revolutionized the way students connect with peers, share information, and express themselves online.

The popularity of social networking sites began to soar around the mid-2000s, with Facebook emerging as a dominant player in the market. Over the years, the landscape has evolved, with new platforms and features continuously being introduced. This rapid growth has had a profound impact on student life, influencing their social interactions, academic pursuits, and overall well-being.

The outbreak of the COVID-19 pandemic further amplified the role of social networking sites in students' lives, as quarantine and social distancing measures forced them to rely more on these platforms for socializing and staying connected with friends and classmates. This increased reliance on social networking sites has led to changes in how students manage their time, with many spending more hours online than before.

As a result, we have witnessed a significant shift in students' social habits and behaviors, with social networking sites playing a central role in shaping their identities and relationships. Understanding these changes is crucial for educators, parents, and policymakers to support students in navigating the digital landscape and ensuring a healthy balance between online and offline interactions.

### B. Research Questions:

- RQ general:
  - (a) General information (age, sex etc).
- RQ1:
  - (a) Those who prefer to use social media frequently, what kind of impacts are visible on them?
- RQ2:
  - (a) Does the different types of sites/apps used by students affect daily screen time significantly?
  - (b) Which social media site do students use the most?
  - (c) Do the majority of students have a high daily screen time?
- RQ3:
  - (a) Do students have a greater preference of using social media for entertainment purposes than educational?
- RQ4:
  - (a) Does social media addiction carry a significant negative influence on academic performances of students according to them?
- RQ5:
  - (a) Do high number of students feel distracted from other works due to incessant use of social media?
  - (b) But what percent of them seek to avoid distraction due to social media through restricting its usage?

- RQ6:
  - (a) The noticeable behavioral change in students due to incessant use of social media.

## Data Collection Methods

### A. Plan of enquiry:

Here, as our survey method we have chosen online questionnaire survey because of the benefits.

- It saves our time and resources.
- High reach of circulation unlike door to door survey.

Although we have some cons in questionnaire like

- Our reach is restricted to only those who use mobile and have active internet connection.

Figure 1: Stats regarding social networking sites usage

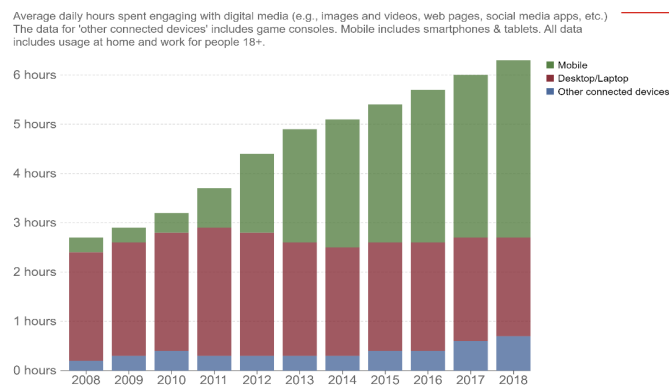


Figure 2: 5G adoption drives 3.6x more usage than 4G in India: Nokia MBit 2024



The impact of 5G on students in India is expected to be significant, providing them with access to faster and more reliable internet connectivity. This could facilitate enhanced learning experiences through online education platforms, improved access to educational resources, and the ability to collaborate on projects remotely. Additionally, 5G could enable new learning technologies such as augmented reality (AR) and virtual reality (VR), creating more immersive and interactive educational experiences. However, challenges such as ensuring equitable access to 5G-enabled devices and addressing potential distractions from increased connectivity may also need to be addressed.

Hence we can clearly see that the impact of 5G is huge among the students and in the same time we want to examine or wanted to know the impact of internet usage among students. Here we want to collect the data about the preference of the students on which platform they prefer to spend time and also the many things like is there any relationship between their sleep hours, is there any impact for excessive use of internet any many more.

The first part of the questionnaire included questions regarding personal information of the respondents, like their age, gender, profession, and where do they live. The second part of the questionnaire included questions regarding the usage like how many hours they use internet and favorite platform, why they use the internet for and many more. The third part of the questionnaire included some basic behavior of a student like do they like to read? what's your opinion about politics and are they recognized themselves as an introvert or extrovert etc.

## B. Target Population and Sampling Frame

Our target population was the people living in Kolkata and its surrounding places and also villages, as we are interested to know the change in their internet usage habits in recent times.

## C. Sampling Scheme

Looking at the data we were able to gather from the survey, we realized that our responses were filled with a majority of respondents belonging to the age-group of 18-25 years. As a result of which, we decided to proceed with simple random sampling (SRS) mechanism (where each element in the population has an equal probability of being selected to the sample).

## D. Design of the Questionnaire

To design the Questionnaire we followed this steps:

- Introduction: Start with an introduction explaining the purpose of the study and ensuring confidentiality and voluntary participation.
- 1. General Information (RQ general): Age: \_\_\_\_\_ & Gender: Male / Female / Other / Prefer not to say
- 2. Educational Level: High School / College / Postgraduate Frequency
- 3. Impact of Social Media Use (RQ1): How frequently do you use social media? (Multiple choice: Several times a day / Once a day / A few times a week / Rarely / Never)
- 4. How do you think frequent use of social media impacts your daily life? (Open-ended)
- 5. Types of Social Media Used and Screen Time (RQ2): Which social media platforms do you use regularly? (Multiple choice: Facebook, Instagram, Twitter, Snapchat, etc.)
- 6. How many hours per day do you spend on social media? (Open-ended)
- 7. Purpose of Social Media Use (RQ3): Do you use social media more for entertainment or educational purposes? (Multiple choice: Entertainment / Educational / Both equally)
- 8. Social Media Addiction and Academic Performance (RQ4): Do you feel addicted to social media? (Yes / No)
- 9. Do you think social media addiction affects your academic performance? (Yes / No / Not sure)
- 10. Distractions and Coping Mechanisms (RQ5): Do you feel distracted from other activities due to social media use? (Yes / No)
- 11. Behavioral Changes (RQ6): Have you noticed any behavioral changes in yourself due to excessive social media use? (Open-ended)
- Conclusion: Thank the participants for their time and provide contact information for any questions or concerns. Optional: Include a section for additional comments or feedback.

Although, we faced several problems at first like the correct order of the questions and actually what kind of questions are valid and etc.

## E. Follow-up of the process

During the initial stages of data collection, we observed that a major portion of the recorded responses was from the age group of 14-30 years which was quite obvious as young adults are generally more accustomed to these kind of services than the elderly section of the population. Hence we tried to circulate the google form among people falling in those age groups. We also asked everyone to re-share our questionnaire, especially among those people in their contacts belonging to our required age group, so that our reach and range of data increases.

Figure 3: Provisional 1st Questionnaire

| <u>Topic</u>                                                                                                                                                                                                                                                                               |  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| The impact of social networking sites on the studies of students near Kolkata                                                                                                                                                                                                              |  |
| <b>Provisional Questionnaire: (Manish &amp; Sourav Dey)</b>                                                                                                                                                                                                                                |  |
| Dear respondent,                                                                                                                                                                                                                                                                           |  |
| We are students of Presidency University working on a survey report about the “The impact of social networking sites on the studies of students near Kolkata”. Please fill up the questionnaire the information given by you will be kept confidential and used for research purpose only. |  |
| <b>(Sourav Dey)</b>                                                                                                                                                                                                                                                                        |  |
| <b>Part-A</b>                                                                                                                                                                                                                                                                              |  |
| 1. Name:                                                                                                                                                                                                                                                                                   |  |
| 2. Age:                                                                                                                                                                                                                                                                                    |  |
| 3. Sex:                                                                                                                                                                                                                                                                                    |  |
| 4. Educational Qualification <b>(Sourav Dey)</b>                                                                                                                                                                                                                                           |  |
| i) UG                                                                                                                                                                                                                                                                                      |  |
| ii) PG                                                                                                                                                                                                                                                                                     |  |
| iii) Graduated                                                                                                                                                                                                                                                                             |  |
| iv) Profession                                                                                                                                                                                                                                                                             |  |
| 5. Do you use social networking sites? <b>(AKG Sir)</b>                                                                                                                                                                                                                                    |  |
| i) Yes                                                                                                                                                                                                                                                                                     |  |
| ii) No                                                                                                                                                                                                                                                                                     |  |
| [ If the answer of the above question is “No”. Then you don’t need to fill the following!]                                                                                                                                                                                                 |  |
| 6. How often do you use social networking sites? <b>(Sourav Dey)</b>                                                                                                                                                                                                                       |  |
| i) Not at all                                                                                                                                                                                                                                                                              |  |
| ii) Rarely                                                                                                                                                                                                                                                                                 |  |
| iii) Infrequently                                                                                                                                                                                                                                                                          |  |
| iv) Frequently                                                                                                                                                                                                                                                                             |  |
| v) Always                                                                                                                                                                                                                                                                                  |  |
| 7. How long hours a day do you spend on social networking websites? <b>(Modified version of Manish by Sourav Dey)</b>                                                                                                                                                                      |  |
| i) 0-1 hours                                                                                                                                                                                                                                                                               |  |
| ii) 1-2 hours                                                                                                                                                                                                                                                                              |  |
| iii) 2-3 hours                                                                                                                                                                                                                                                                             |  |
| iv) 3-5 hours                                                                                                                                                                                                                                                                              |  |
| v) 5+ hours                                                                                                                                                                                                                                                                                |  |
| 8. If you are college student which stream you’re in? <b>(Manish)</b>                                                                                                                                                                                                                      |  |
| i) Sc                                                                                                                                                                                                                                                                                      |  |
| ii) Arts                                                                                                                                                                                                                                                                                   |  |
| 9. Do you like to read books as daily basis? <b>(AKG Sir)</b>                                                                                                                                                                                                                              |  |
| i) Yes                                                                                                                                                                                                                                                                                     |  |
| ii) No                                                                                                                                                                                                                                                                                     |  |
| 10. Daily data usage: (*in Giga Bites*) <b>(Manish)</b>                                                                                                                                                                                                                                    |  |
| i) 1-1.5                                                                                                                                                                                                                                                                                   |  |
| ii) 1.5-2.5                                                                                                                                                                                                                                                                                |  |
| iii) 2.5-5                                                                                                                                                                                                                                                                                 |  |
| iv) 5+                                                                                                                                                                                                                                                                                     |  |
| 11. Favorite Social networking platform to spend time: <b>(Manish)</b>                                                                                                                                                                                                                     |  |
| i) Instagram                                                                                                                                                                                                                                                                               |  |
| ii) Facebook                                                                                                                                                                                                                                                                               |  |
| iii) YT                                                                                                                                                                                                                                                                                    |  |
| iv) Chrome                                                                                                                                                                                                                                                                                 |  |
| v) Telegram                                                                                                                                                                                                                                                                                |  |
| vi) X                                                                                                                                                                                                                                                                                      |  |
| vii) Others                                                                                                                                                                                                                                                                                |  |
| 12. Why do you spend time on social networking sites? <b>(Modified version of Manish by Sourav Dey)</b>                                                                                                                                                                                    |  |
| i) Chatting & relaxing                                                                                                                                                                                                                                                                     |  |
| ii) For study                                                                                                                                                                                                                                                                              |  |
| iii) Writing blogs                                                                                                                                                                                                                                                                         |  |
| iv) Others                                                                                                                                                                                                                                                                                 |  |
| 13. What do you prefer/ Which one do you prefer? <b>(Manish)</b>                                                                                                                                                                                                                           |  |
| i) Offline classes                                                                                                                                                                                                                                                                         |  |
| ii) Online classes                                                                                                                                                                                                                                                                         |  |
| 14. Do you feel stressed out for over usage of social networking sites. <b>(Manish)</b>                                                                                                                                                                                                    |  |
| i) Yes                                                                                                                                                                                                                                                                                     |  |
| ii) No                                                                                                                                                                                                                                                                                     |  |
| 15. Does this effect your life? <b>(Manish)</b>                                                                                                                                                                                                                                            |  |
| i) Yes                                                                                                                                                                                                                                                                                     |  |
| ii) No                                                                                                                                                                                                                                                                                     |  |
| 16. Money spends on the internet in each month: (*in rupees*) <b>(Manish)</b>                                                                                                                                                                                                              |  |
| i) 0-250                                                                                                                                                                                                                                                                                   |  |
| ii) 250-500                                                                                                                                                                                                                                                                                |  |
| iii) 500-1000                                                                                                                                                                                                                                                                              |  |
| iv) 1000+                                                                                                                                                                                                                                                                                  |  |

Figure 4: Final Questionnaire



## Unlock the insights

Title: **The impact of social networking sites on students around Kolkata.**

Dear Participant,

Thank you for agreeing to participate in this research study aimed at understanding the impact of social media on students' academic performance. Your insights are invaluable in helping us comprehend the complexities of this issue.

This questionnaire is designed to gather information about your social media usage patterns, academic performance, and perceptions regarding the influence of social media on your studies. Your responses will be kept confidential and used solely for research purposes.

Please answer each question honestly and to the best of your ability. There are no right or wrong answers, and your participation is entirely voluntary. If you feel uncomfortable answering any question, you may skip it.

Your cooperation is greatly appreciated, and your contribution will contribute to the advancement of knowledge in this field.

Thank you for your time and participation.

Sincerely,  
[ Swarnendu Dey, Sourav Dey, Manish Chandra Roy, Roshni Majhi ]  
[ Department of Statistics, Presidency University ]

dey10752@gmail.com [Switch accounts](#)

Not shared

**Name: \***

Your answer

**Sex \***

☐ Male

☐ Female

☐ Prefer not to say

☐ Other: \_\_\_\_\_

**Age: \***

Your answer

**Annual family income : \***

☐ Below 60000

☐ 60,000 - 3,40,000

☐ 3,40,000 - 7,80,000

☐ 7,80,000 - 10,00,000

☐ Above 10,00,000

**Where do you stay? \***

☐ Kolkata

☐ Surroundings

☐ Villages

☐ Other: \_\_\_\_\_

**Average regular commute hours to go to your college or university. \***

Your answer

**How often do you use social networking sites? \***

☐ Not at all

☐ Rarely

☐ Moderately

☐ Frequently

☐ Always

**How many hours a day do you spend time on social networking websites on an average? \***

Your answer

**Do you enjoy reading? \***

☐ Yes

☐ No

**Which language do you prefer for reading books? \***

☐ Bengali

☐ English

☐ None

☐ Other: \_\_\_\_\_

**In which stream are you pursuing your bachelors/post graduate/PhD? \***

☐ Humanity

☐ Science

☐ Commerce

☐ Technical

☐ Medical or medical related any course

☐ Other: \_\_\_\_\_

**Which one is your preferred genre? \***

☐ Science fiction

☐ Narrative

☐ Novel

☐ Mystery

☐ None

☐ Other: \_\_\_\_\_

**Favorite Social networking platform to spend time on: \***

☐ whatsapp

☐ Instagram

☐ Facebook

☐ Youtube

☐ Telegram

☐ twitter

☐ None

☐ Other: \_\_\_\_\_

**Do you feel stressed out for using too much of social networking sites? \***

☐ Yes

☐ No

**In your opinion how would you rate your academic satisfaction out of 5. \***

☐ 1

☐ 2

☐ 3

☐ 4

☐ 5

☐ Not at all

**What is the significant purpose of yours to use social networking sites? \***

☐ Writing blogs

☐ Gathering information

☐ Studying

☐ Chatting and relaxing

☐ Other: \_\_\_\_\_

**In your opinion how would you rate your professional satisfaction out of 5, professional satisfaction \***

☐ 1

☐ 2

☐ 3

☐ 4

☐ 5

☐ Not at all

**From where do you get to know the socio- political scenario of recent times? \***

☐ Newspapers

☐ news channels

☐ Social networking sites

☐ Not interested at all

☐ Other: \_\_\_\_\_

**How would you find yourself in a social gathering? \***

☐ Extroverted

☐ Introverted

## F. Data description :

```
'data.frame': 140 obs. of 22 variables:
 $ Q1 : chr  "2024/04/09 6:24:48 pm GMT+5:30" "2024/04/09 6:33:21 pm GMT+5:30" "2024/04/09 6:52:58 pm GMT+5:30" ...
 $ Q2 : chr  "Sangbartta Banerjee " "Pratik Majumdar " "Upasak Chatterjee " "Sneha Kundu " ...
 $ Q3 : chr  "Male" "Male" "High-Elf" "Female" ...
 $ Q4 : int  19 19 20 19 20 21 19 24 20 20 ...
 $ Q5 : chr  "60,000 - 3,40,000" "60,000 - 3,40,000" "Below 60000" "3,40,000 - 7,80,000" ...
 $ Q6 : chr  "Surroundings" "Kolkata" "Kolkata" "Kolkata" ...
 $ Q7 : chr  "1.5" "1" "less than 30 minutes " "less than 30 minutes " ...
 $ Q8 : chr  "Frequently" "Moderately" "Not at all" "Frequently" ...
 $ Q9 : num  3.5 2 5 3 3 2 4 6 6 3 ...
 $ Q10: chr  "Science" "Science" "Science" "Science" ...
 $ Q11: chr  "Yes" "Yes" "No" "Yes" ...
 $ Q12: chr  "English" "English" "Python" "English" ...
 $ Q13: chr  "Narrative" "Sf" "Cooking" "Mystery" ...
 $ Q14: chr  "YouTube" "Others" "Others" "Instagram" ...
 $ Q15: chr  "Chatting and relaxing" "Chatting and relaxing" "Others" "Chatting and relaxing" ...
 $ Q16: chr  "Social networking sites" "Social networking sites" "Not interested at all" "Social networking sites" ...
 $ Q17: chr  "No" "No" "Yes" "No" ...
 $ Q18: chr  "Yes" "Yes" "No" "Yes" ...
 $ Q19: int  4 4 0 3 3 0 3 0 3 3 ...
 $ Q20: int  4 4 5 3 3 0 4 0 3 3 ...
 $ Q21: chr  "Introverted" "Introverted" "Introverted" "Extroverted" ...
 $ Q22: num  7 7 6 6 7 5 6 6 6 9 ...
```

Here after collecting the data, we got the following variables w.r.t. Q1,Q2, and so on respectively:

- Q1: Time
- Q2: Name
- Q3: Gender
- Q4: Age
- Q5: Family income
- Q6: Locality
- Q7: Commute hours to go to their college or university
- Q8: How often do they use social networking sites
- Q9: Time spending on the internet
- Q10: Stream
- Q11: Do they enjoy reading?
- Q12: Language they prefer for reading
- Q13: Favorite genre
- Q14: Favorite platform
- Q15: Significant purpose of yours to use social networking sites
- Q16: Where do you get to know the socio - political scenario
- Q17: Subscription of any online education course
- Q18: Do you feel stressed out for using too much of social networking sites
- Q19: Academic satisfaction out of 5
- Q20: Professional satisfaction out of 5
- Q21: How would you find yourself in a social gathering
- Q22: Sleep hours

# Statistical methodology

## A. Scrutiny of Data

After collecting the data we found a lot of information which were incomplete in the disguise of missing. 5 shows us a few examples of data which were incorrect when cross verified. These data were eliminated after complete case analysis, otherwise these could've affected the summary of the variable behaving as outliers.

Figure 5: Examples of Incorrect/contradictory data

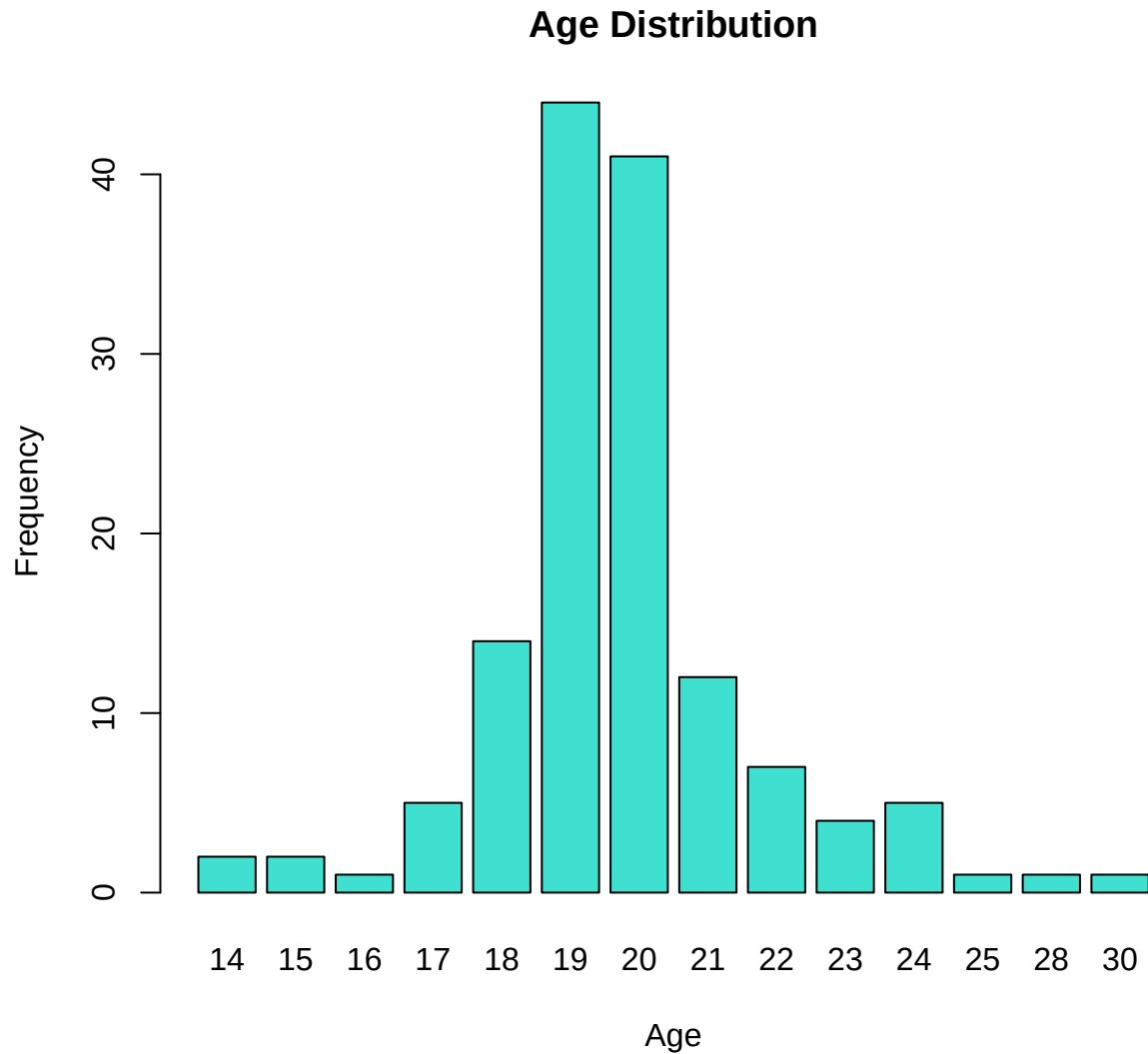
|                                                                    |                                      |
|--------------------------------------------------------------------|--------------------------------------|
| Sex                                                                | • High-Elf                           |
| Age:                                                               | • 1213 years                         |
| Average regular commute hours to go to your college or university. | • 7 days walking then 2 days cycling |
| How many hours do you sleep in a day?                              | • 2minutes 22 secs                   |

## B. Variable Pre processing :

Many of the respondents answered the Question: “How many hours a day do you spend time on social networking websites on an average?” as a range like 3-4, 5-6 etc. As we asked value as an average, we replaced the given range as the mean of the two values given.

## C. Exploratory Data Analysis

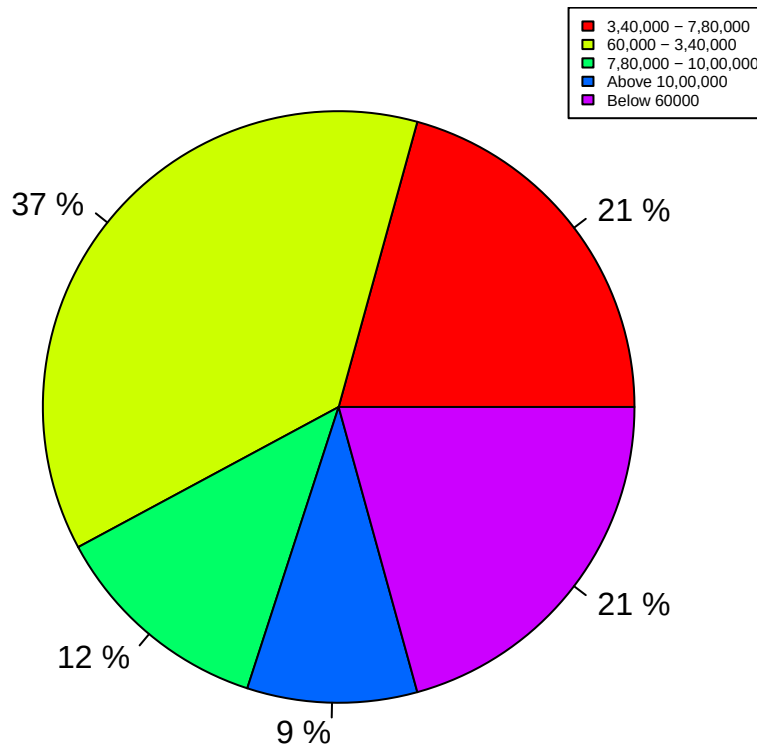
We start by showing the Column diagram (bar plot) of the age distribution & pie chart for annual family income from where we were collected the data.



This histogram shows that our sample is largely composed of individuals aged 18 to 21, with the peak frequencies at ages 19 and 20. This demographic concentration is significant because it suggests our findings will primarily reflect the behaviors and attitudes of late teenagers and young adults.

The smaller number of respondents in the younger and older age brackets indicates that our data may be less representative of those age groups. However, the central cluster provides a robust basis for understanding the impact of social networking sites on typical college-age students.

### Annual Family Income Distribution



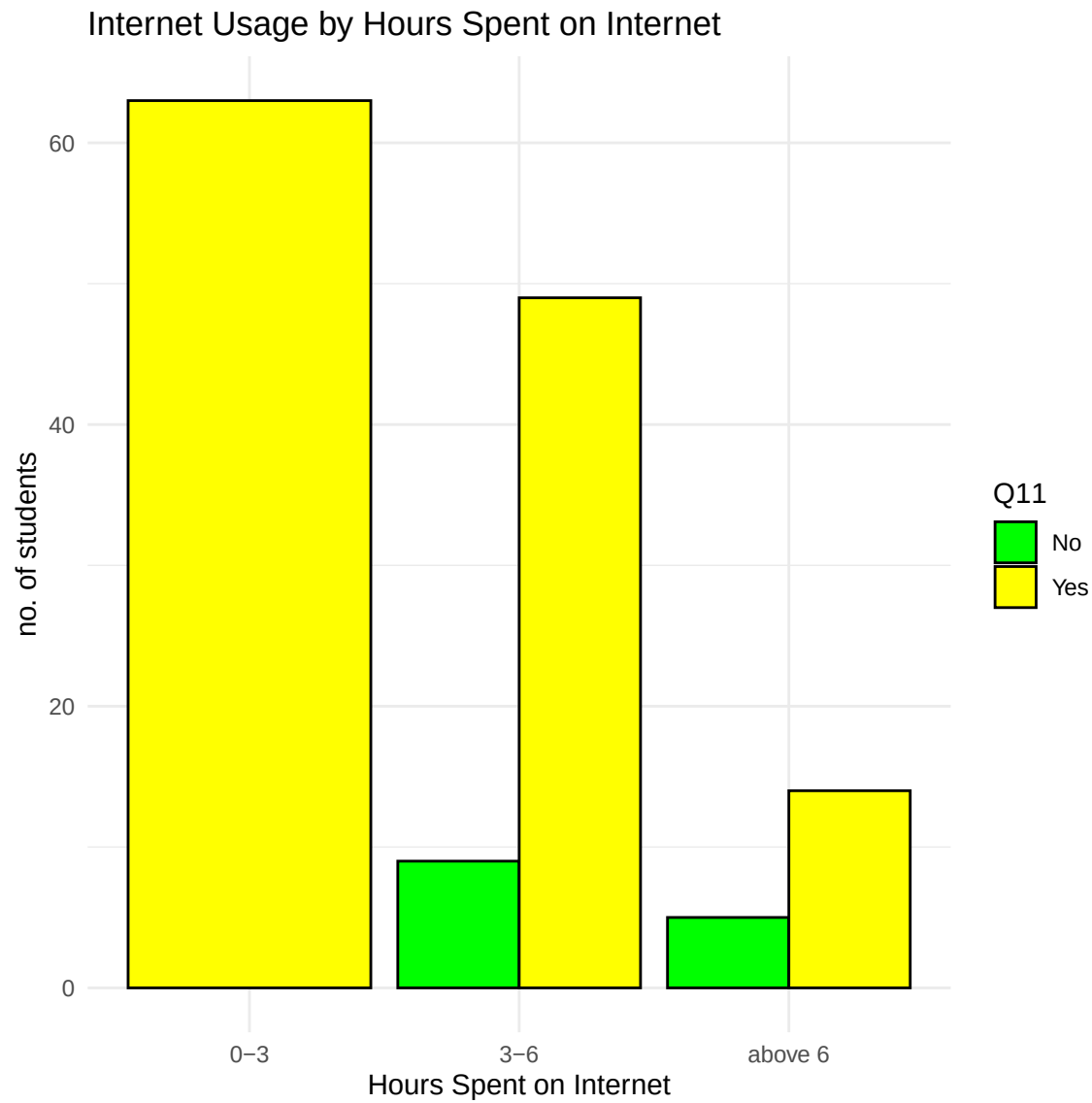
This pie chart illustrates that a significant portion of our respondents, 37%, fall within the ₹7,80,000 – ₹10,00,000 income range. This is the most common income bracket among our participants.

Approximately 21% of respondents have a family income between ₹3,40,000 and ₹7,80,000, and another 21% fall in the ₹60,000 – ₹3,40,000 range. These two middle-income brackets together account for a substantial portion of our sample.

Respondents with an annual family income above ₹10,00,000 constitute 12% of the sample, while those with an income below ₹60,000 make up the smallest group at 9%.

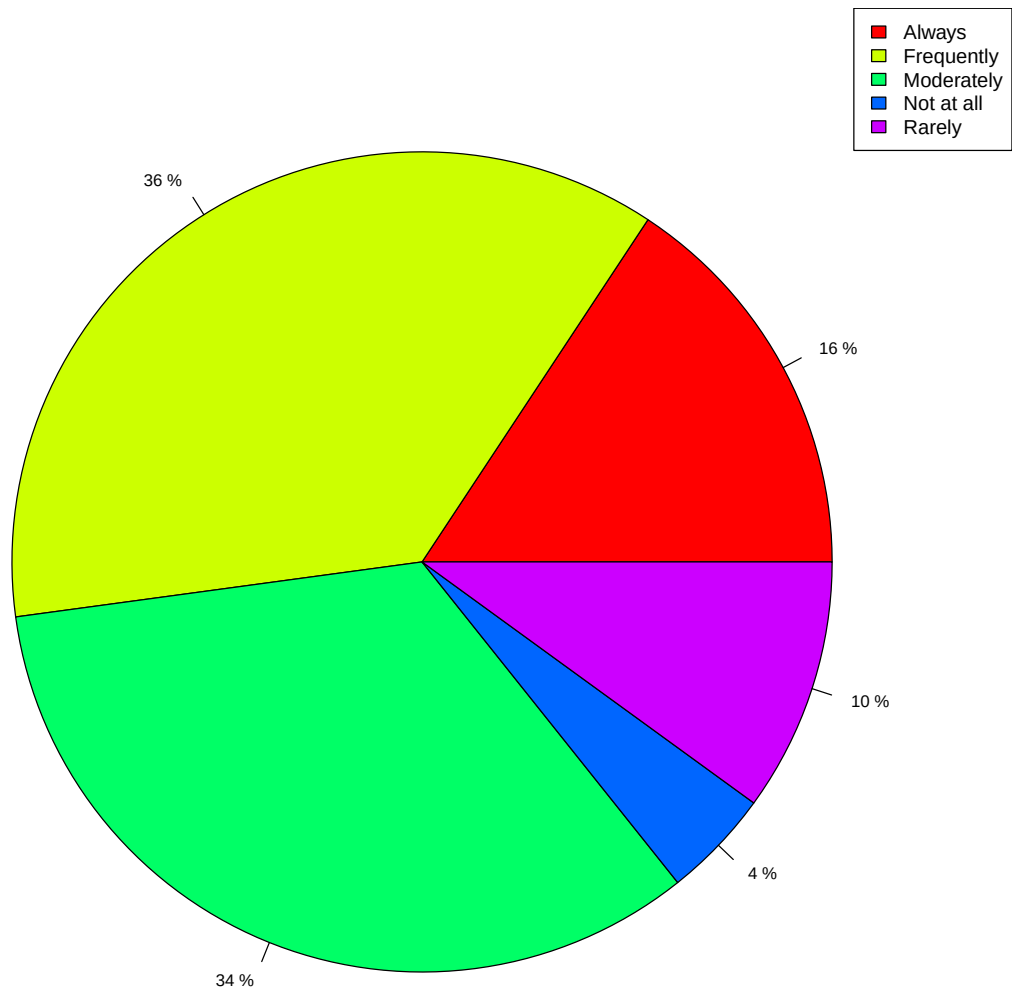
These income distributions help us understand the socio-economic backgrounds of our respondents, which is crucial for analyzing how financial status might influence social media usage and its impact.

Warning: package 'ggplot2' was built under R version 4.3.3



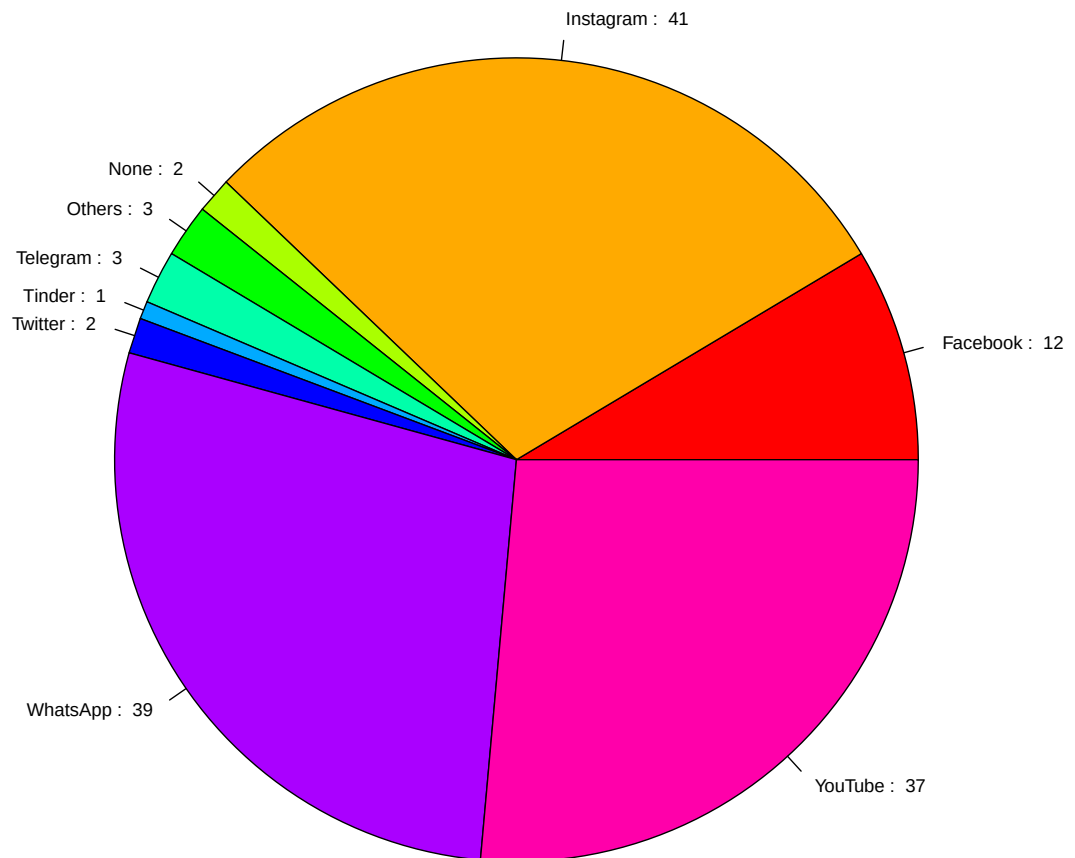
Another interesting fact we can observe based on students reading behavior, we asked students whether they like to read or not? And we got some interesting results at the end, fact or the data showing us that there is no dependency when it comes to internet usage time of a student and reading behavior, most of the students like to read books although they spend many times on the internet.

## Social Networking Usage Distribution



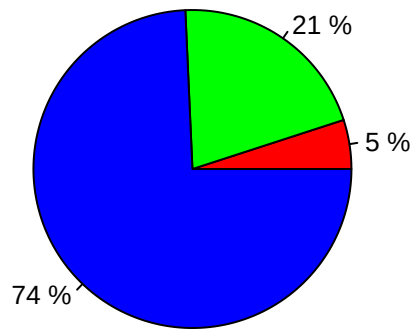
While collecting the data, we were also interested in knowing how often these people use social media/social networking sites. We're not surprised seeing that almost 16% of the population prefer to use social networking websites in a daily basis and they quite addicted to it. Although almost 36% and 34% of the people still prefer to use social networking sites frequently and moderate respectively by not adopting the new habits of these generation.

## Favorite Social Networking Platform

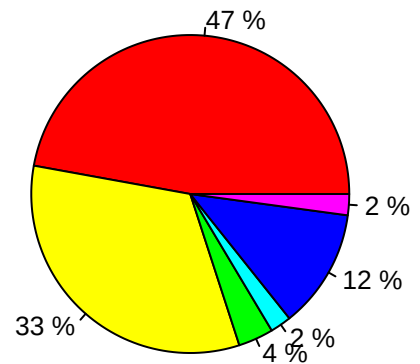
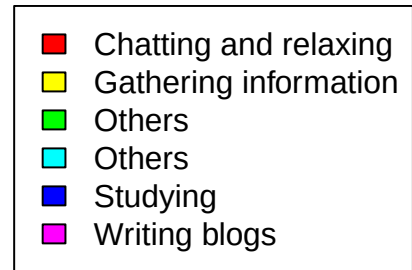


We collected the names of social networking sites/apps people used the most, and considered multiple names, and did a pie chart. Instagram emerged as the most used app, followed by Whats app and YouTube respectively.

**Stream of Study Distribution**

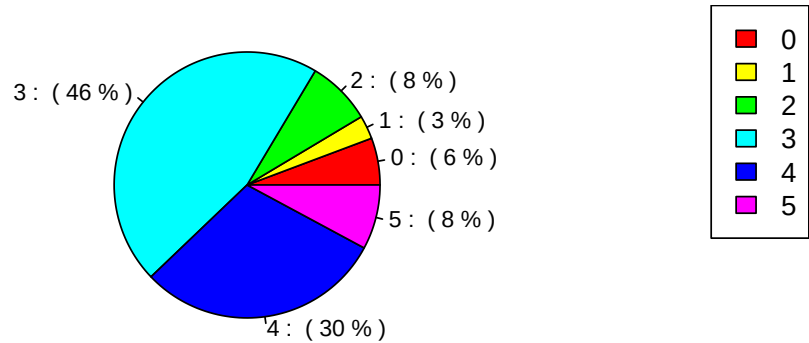


**Purpose of Using Social Networking Sites**

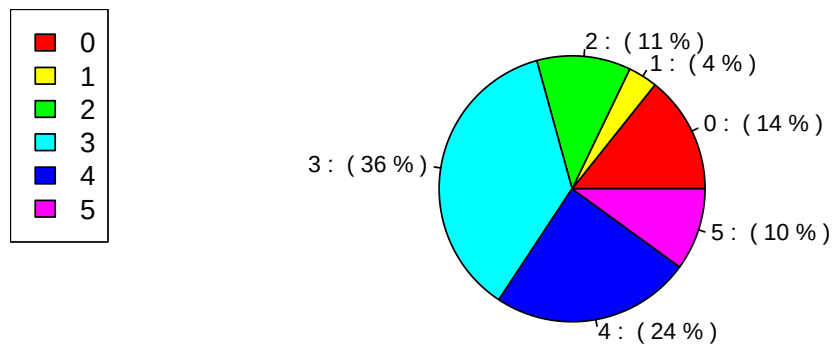


We were also interested to find the purpose of usage of these social networking websites. As we collected the data mostly from the students we did two pie charts about in which stream they are pursuing in and why do they use social networking sites in their daily life and found that 76% of the students are pursuing science and among all 49% of the students use social networking websites for chatting and relaxing, then comes the reason gathering information.

### Academic Satisfaction Rating

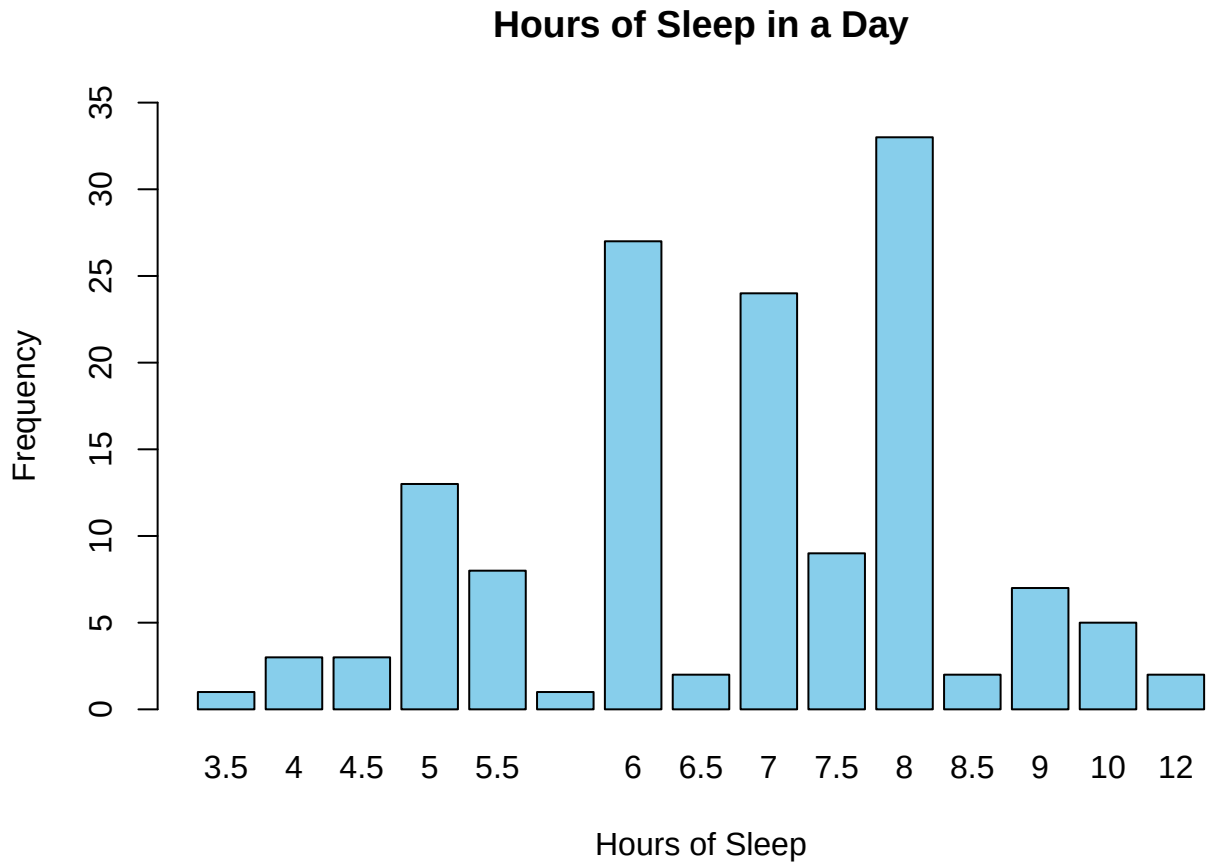


### Professional Satisfaction Rating



A pie chart on the academic satisfaction out of 5, next is professional satisfaction out of 5 of all the respondents. The data is represented academic satisfaction ratings show a significant proportion at level 3 (red) and a substantial group at level 4 (light blue). Professional satisfaction ratings highlight a dominant level 3 (red) and a smaller group at level 4 (light blue).

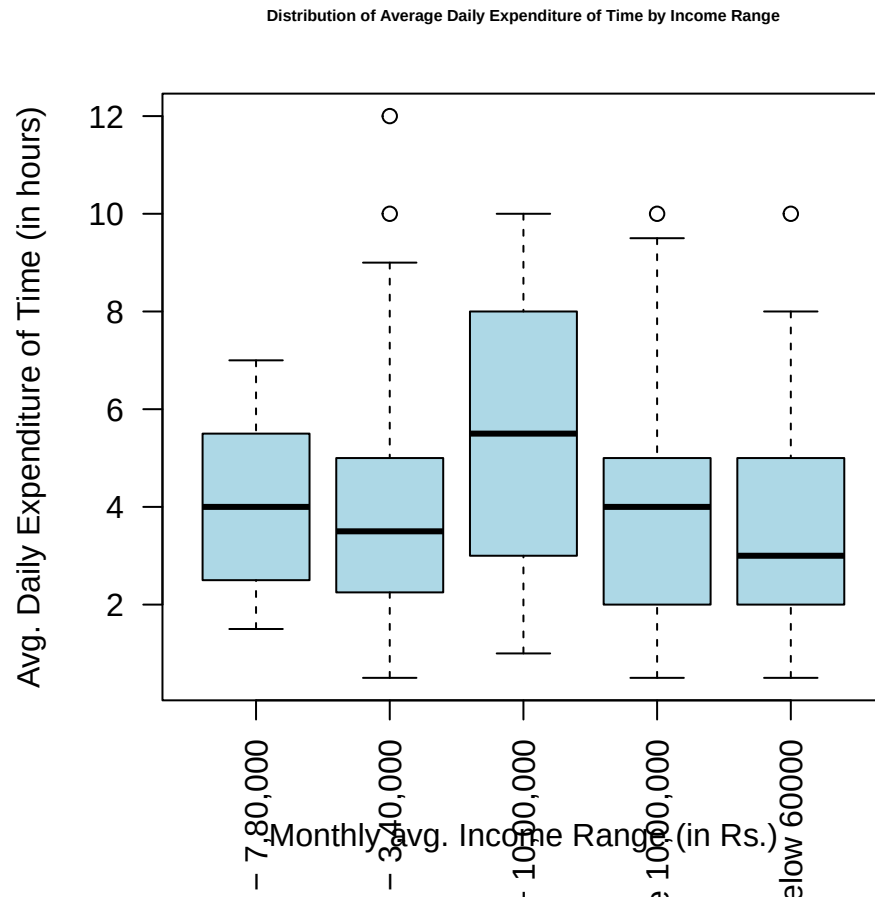
- The majority of youth feel moderately satisfied in both overall and professional contexts.
- There's a higher percentage of youth giving low satisfaction ratings (0-2) in professional satisfaction compared to overall satisfaction.
- Positive Sentiment: A significant portion of the youth still rate their satisfaction highly (4-5) in both categories.



- The data indicates that most individuals get between 6 to 8 hours of sleep, which aligns with general sleep recommendations for young adults.
- There are fewer individuals getting extremely low (less than 5 hours) or extremely high (more than 9 hours) amounts of sleep.

The peak at 8 hours suggests that a significant portion of the group is likely meeting the recommended amount of sleep, which is beneficial for health and well-being. The histogram provides a clear visual representation of sleep patterns, showing that 6 to 8 hours of sleep is the most common duration among the group surveyed.

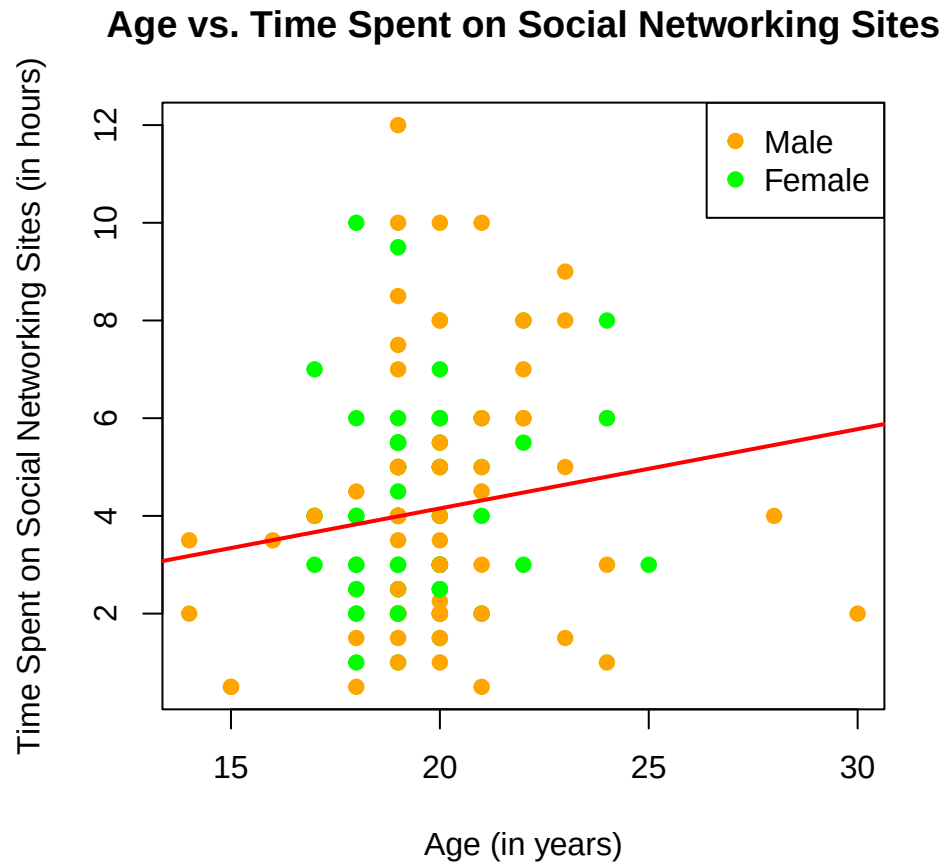
It is essential to encourage maintaining healthy sleep habits, aiming for 7-8 hours per night.



Observations:

1. People with higher incomes tend to spend more time daily.
2. Income groups with wider boxes have more variation in time use.
3. Outliers (unusual cases) can impact the overall pattern.

In summary, this graph helps us understand how income influences daily time allocation.



We also tried to fit a regression to predict the average time spend on social networking sites given the age of the students, which although wasn't quite good.

Warning: package 'vcd' was built under R version 4.3.3

Loading required package: grid

[1] 0.2428948

There seems to be a weak to moderate association between age and time spent on social networking sites.

## D. Inferential Data Analysis

- For the categorical variables we have we decided to perform a chi-square test of independence.

Hypothesis :

we want to test  $H_0$  :their is no relation or avg. number of hours they spend on their internet does not any effect in their academic satisfaction level

against the alternative hypothesis  $H_1$  : they are related

we divide the avg. spending time on the internet into three groups "0-3 hours" , "3-6 hours " and above 6 hours

```

                                academic_satisfaction_level(0-5)
avg_hour_they_spend_on_the_internet 0-2 3  4-5
                                0-3 hours    7  30 26
                                3-6 hours    8  29 21
                                above 6 hours 8   5  6

```

```
Warning in chisq.test(cont_table): Chi-squared approximation may be incorrect
```

Pearson's Chi-squared test

```
data: cont_table
X-squared = 11.235, df = 4, p-value = 0.02405
```

Since, we see that the p-value of the chi-square test of independence is less than the level of significance (0.05) so in the light of the given data, we do not have sufficient reason to believe that the variables are not related, that is, we reject the null hypothesis, which states that avg. number of hours they spend on their internet does effect in their academic satisfaction level.

- Another test we can do is to check whether there is a relation between the stream they are in and their avg. time spend on their social networking sites.

Hypothesis :

we want to test  $H_0$  :their is no relation between the stream they are in and their avg. time spend on their social networking sites.

against the alternative hypothesis  $H_1$  : they are related

```
[1] "numeric"
[1] "character"
      0-3 3-6 above 6
SCI/THECH/MED      46  47      11
HUM/MBA/COMM/O.TW  17  11       8
```

```
Warning in chisq.test(obs.freq, correct = T): Chi-squared approximation may be incorrect
```

Pearson's Chi-squared test

```
data: obs.freq
X-squared = 4.1084, df = 2, p-value = 0.1282
```

Since, we see that the p-value of the chi-square test of independence is more than the level of significance(0.05) so in the light of the given data, we do have sufficient reason to believe that the variables are not related, that is we accept the null hypothesis, which states that there is no relation between the stream they are in and their avg. time spend on their social networking sites.

- Now for the another example we are going to check is there any relation between a student's locality and his/her internet usage hours ?

Hypothesis :

we want to test  $H_0$  :their is no relation between a student's locality and his/her internet usage hours .

against the alternative hypothesis  $H_1$  : they are related

```
[1] "numeric"
[1] "character"
      0-3 above 3 hours
Surroundings  22      21
Kolkata       34      49
Villages       7       7
```

Pearson's Chi-squared test

```
data: obs.freq
X-squared = 1.3476, df = 2, p-value = 0.5098
```

Since, we see that the p-value of the chi-square test of independence is more than the level of significance (0.05) so in the light of the given data, we do have sufficient reason to believe that the variables are not related, that is, we accept the null hypothesis, which states that there is no relation between a student's locality and his/her internet usage hours.

- Another example

Hypothesis :

we want to test  $H_0$  :there is no relation between the style or the reason for their internet usage and the stream they choose.

against the alternative hypothesis  $H_1$  : they are related

```
[1] "character"
[1] "character"
               SCI/THECH/MED HUM/MBA/COMM/O.TW
Chatting and relaxing           51             15
Gathering information           34             12
studying                        12              5

Warning in chisq.test(obs.freq_Q15_Q10):  Chi-squared approximation may be incorrect

Pearson's Chi-squared test

data: obs.freq_Q15_Q10
X-squared = 0.3867, df = 2, p-value = 0.8242
```

Since, we see that the p-value of the chi-square test of independence is more than the level of significance (0.05) so in the light of the given data, we do have sufficient reason to believe that the variables are not related, that is, we accept the null hypothesis, which states that there is no relation between the style or the reason for their internet usage and the stream they choose.

## Appendix

**R codes:**

**Data Collection Methods**

**F. Data description :**

```
My.file=read.csv("C:\\Users\\SOURAV\\Desktop\\SEM - 4\\SEM - 4 AKG\\Group Report Survey\\Unlock the insight
colnames(My.file)=paste("Q",1:ncol(My.file),sep="")
str(My.file)
```

## Statistical methodology

### C. Exploratory Data Analysis

Column diagram (bar plot) of the age distribution & pie chart for annual family income.

```
barplot(table(My.file$Q4),
        main = "Age Distribution",
        xlab = "Age",
        ylab = "Frequency",
        col = "turquoise")
```

```
income_counts <- table(My.file$Q5)
pie(income_counts, labels = paste(round(income_counts/sum(income_counts)*100), "%"),
    main = "Annual Family Income Distribution", cex.main=0.9,
    col = rainbow(length(income_counts)))
legend("topright", cex=0.48, legend = names(income_counts),
    fill = rainbow(length(income_counts)))
```

Column diagram on Internet Usage by Hours Spent on Internet.

```
My.file$Q9_group <- cut(My.file$Q9,breaks <- c(0,3,6,14), labels = c("0-3", "3-6","above 6"), include.lowest=TRUE)
library(ggplot2)
ggplot(My.file, aes(x = Q9_group, fill = Q11)) +
  geom_bar(position = position_dodge(width = 0.9), color = "black", stat = "count") +
  labs(title = "Internet Usage by Hours Spent on Internet", x = "Hours Spent on Internet", y = "no. of users") +
  scale_fill_manual(values = c("No" = "green", "Yes" = "yellow")) +
  theme_minimal()
```

Pie chart on Social Networking Usage Distribution.

```
social_counts <- table(My.file$Q8)
pie(social_counts,
    labels = paste(round(social_counts/sum(social_counts)*100), "%"),
    main = "Social Networking Usage Distribution",
    cex.main=2,
    col = rainbow(length(social_counts)),
    cex = 0.8)
legend("topright", cex=0.98, legend = names(social_counts), fill = rainbow(length(social_counts)))
```

Pie chart on Favorite Social Networking Platform.

```
platform_counts <- table(My.file$Q14)
pie(platform_counts,
    labels = paste(names(platform_counts), ": ", platform_counts),
    main = "Favorite Social Networking Platform",
    cex.main=2,
    col = rainbow(length(platform_counts)),
    cex = 0.8)
```

Pie chart on Stream of Study Distribution and pie chart on Purpose of Using Social Networking Sites.

```
par(mfrow=c(1,2))
stream_counts <- table(My.file$Q10)
pie(stream_counts,
    labels = paste(round(stream_counts/sum(stream_counts)*100), "%"),
    main = "Stream of Study Distribution", cex.main=0.9,
    col = rainbow(length(stream_counts)),
    cex = 0.8)
```

```

legend("topleft", legend = names(stream_counts), fill = rainbow(length(stream_counts)))

purpose_counts <- table(My.file$Q15)
pie(purpose_counts,
    labels = paste(round(purpose_counts/sum(purpose_counts)*100), "%"),
    main = "Purpose of Using Social Networking Sites", cex.main=0.9,
    col = rainbow(length(purpose_counts)),
    cex = 0.8)
legend("topright", legend = names(purpose_counts), fill = rainbow(length(purpose_counts)))

```

Pie chart on academic satisfaction and professional satisfaction.

```

par(mfrow=c(2,1))
satisfaction_counts <- table(My.file$Q19)
pie(satisfaction_counts,
    labels = paste(names(satisfaction_counts), ": ", "(",
    round(satisfaction_counts/sum(satisfaction_counts)*100), "%", ")"),
    main = "Academic Satisfaction Rating", cex.main=0.9,
    col = rainbow(length(satisfaction_counts)),
    cex = 0.8)
legend("topright", legend = names(satisfaction_counts), fill = rainbow(length(satisfaction_counts)))

satisfaction_counts_1 <- table(My.file$Q20)
pie(satisfaction_counts_1,
    labels = paste(names(satisfaction_counts_1), ": ",
    "(", round(satisfaction_counts_1/sum(satisfaction_counts_1)*100), "%", ")"),
    main = "Professional Satisfaction Rating", cex.main=0.9,
    col = rainbow(length(satisfaction_counts_1)),
    cex = 0.8)
legend("topleft", legend = names(satisfaction_counts_1), fill = rainbow(length(satisfaction_counts_1)))

```

Column diagram on Hours of Sleep in a Day.

```

sleep_counts <- table(My.file$Q22)
barplot(sleep_counts,
    main = "Hours of Sleep in a Day",
    xlab = "Hours of Sleep",
    ylab = "Frequency",
    col = "skyblue",
    ylim = c(0, max(sleep_counts) + 2))

```

Box plot:

```

My.file$Q5 <- as.factor(My.file$Q5)
My.file$Q9 <- as.numeric(My.file$Q9)
boxplot(Q9 ~ Q5,
    data = My.file,
    main = "Distribution of Average Daily Expenditure of Time by Income Range",
    xlab = "Monthly avg. Income Range (in Rs.)",
    ylab = "Avg. Daily Expenditure of Time (in hours)",
    col = "lightblue",
    las=2,
    angle=45,)

```

Fitting regression line.

```
My.file$Q9 <- as.numeric(My.file$Q9)
age <- My.file$Q4
time_spent <- My.file$Q9
gender <- My.file$Q3
plot(age, time_spent,
     main = "Age vs. Time Spent on Social Networking Sites",
     xlab = "Age (in years)",
     ylab = "Time Spent on Social Networking Sites (in hours)",
     pch = 19,
     col = ifelse(gender == "Male", "orange", "green"))
fit <- lm(time_spent ~ age)
abline(fit, col = "red", lwd = 2)
legend("topright", legend = c("Male", "Female"), col = c("orange", "green"), pch = 19)
```

Association test!

```
table_Q6_Q14 <- table(My.file$Q6, My.file$Q14)

library(vcd)

cramers_v <- assocstats(table_Q6_Q14)$cramer

print(cramers_v)
```

## D. Inferential Data Analysis

- 1st Test

```
data1 <- My.file$Q9
breaks <- c(0,3,6,14)
groups <- cut(data1, breaks = breaks, labels = FALSE)
group_labels <- cut(data1, breaks = breaks, labels = c("0-3", "3-6", "above 6 "), include.lowest = TRUE)
data_frame1 <- data.frame(Data = data1, Group = group_labels)

data2 <- My.file$Q19
breaks <- c(-Inf, 2, 3, 5)
groups <- cut(data2, breaks = breaks, labels = FALSE)
group_labels <- cut(data2, breaks = breaks, labels = c("0-2", "3", "4-5"), include.lowest = TRUE)
data_frame2 <- data.frame(Data = data2, Group = group_labels)
data_frame1$Group <- as.factor(data_frame1$Group)
data_frame2$Group <- as.factor(data_frame2$Group)
cont_table <- table(data_frame1$Group, data_frame2$Group)
dimnames(cont_table) <- list(
  "avg_hour_they_spend_on_the_internet" = c("0-3 hours ", "3-6 hours", "above 6 hours"),
  "academic_satisfaction_level(0-5)" = c("0-2", "3", "4-5")
)

print(cont_table, quote = FALSE, right = FALSE)
chi_square_test <- chisq.test(cont_table)
print(chi_square_test)
```

- 2nd Test

```
class(My.file$Q9)
class(My.file$Q10)
My.file$Q9 <- as.numeric(My.file$Q9)
My.file$Q9_group <- cut(My.file$Q9, breaks = c(0, 3, 6, 14), labels = c("0-3", "3-6", "above 6"),
include.lowest = TRUE)
obs.freq <- matrix(0, nrow = 2, ncol = 3)
obs.freq[1, 1] <- sum(My.file$Q10 == "Science" & My.file$Q9_group == "0-3")
obs.freq[1, 2] <- sum(My.file$Q10 == "Science" & My.file$Q9_group == "3-6")
obs.freq[1, 3] <- sum(My.file$Q10 == "Science" & My.file$Q9_group == "above 6")
obs.freq[2, 1] <- sum(My.file$Q10 != "Science" & My.file$Q9_group == "0-3")
obs.freq[2, 2] <- sum(My.file$Q10 != "Science" & My.file$Q9_group == "3-6")
obs.freq[2, 3] <- sum(My.file$Q10 != "Science" & My.file$Q9_group == "above 6")

rownames(obs.freq) <- c("SCI/THECH/MED", "HUM/MBA/COMM/O.TW")
colnames(obs.freq) <- c("0-3", "3-6", "above 6")
obs.freq
chisq.test(obs.freq, correct=T)
```

- 3rd Test

```
class(My.file$Q9)
class(My.file$Q6)
My.file$Q9 <- as.numeric(My.file$Q9)
My.file$Q9_group <- cut(My.file$Q9, breaks = c(0, 3, 6, 14), labels = c("0-3", "3-6", "above 6"),
include.lowest = TRUE)
obs.freq <- matrix(0, nrow = 3, ncol = 2)
obs.freq[1, 1] <- sum(My.file$Q9_group == "0-3" & My.file$Q6 == "Surroundings")
obs.freq[1, 2] <- sum(My.file$Q9_group != "0-3" & My.file$Q6 == "Surroundings")
obs.freq[2, 1] <- sum(My.file$Q9_group == "0-3" & My.file$Q6 == "Kolkata")
obs.freq[2, 2] <- sum(My.file$Q9_group != "0-3" & My.file$Q6 == "Kolkata")
obs.freq[3, 1] <- sum(My.file$Q9_group == "0-3" & My.file$Q6 == "Villages")
obs.freq[3, 2] <- sum(My.file$Q9_group != "0-3" & My.file$Q6 == "Villages")

rownames(obs.freq) <- c("Surroundings", "Kolkata", "Villages")
colnames(obs.freq) <- c("0-3", "above 3 hours")
obs.freq
chi2 <- chisq.test(obs.freq)
chi2
```

- 4th Test

```
class(My.file$Q10)
class(My.file$Q15)
obs.freq_Q15_Q10 <- matrix(0, nrow = 3, ncol = 2)
obs.freq_Q15_Q10[1, 1] <- sum(My.file$Q15 == "Chatting and relaxing" & My.file$Q10 == "Science")
obs.freq_Q15_Q10[1, 2] <- sum(My.file$Q15 == "Chatting and relaxing" & My.file$Q10 != "Science")
obs.freq_Q15_Q10[2, 1] <- sum(My.file$Q15 == "Gathering information" & My.file$Q10 == "Science")
obs.freq_Q15_Q10[2, 2] <- sum(My.file$Q15 == "Gathering information" & My.file$Q10 != "Science")
```

```
obs.freq_Q15_Q10[3, 1] <- sum(My.file$Q15 == "Studying" & My.file$Q10 == "Science")
obs.freq_Q15_Q10[3, 2] <- sum(My.file$Q15 == "Studying" & My.file$Q10 != "Science")

rownames(obs.freq_Q15_Q10) <- c("Chatting and relaxing", "Gathering information", "studying ")
colnames(obs.freq_Q15_Q10) <- c("SCI/TECH/MED", "HUM/MBA/COMM/O.TW")
obs.freq_Q15_Q10
chi2_Q15_Q10 <- chisq.test(obs.freq_Q15_Q10)
chi2_Q15_Q10
```